Population-based Search

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Rethinking Local Search

- Local search strategies are ways designed to solve large complex search problems with small memory.
- A local search strategy maintains and iteratively updates a single candidate solution.
- This resembles an "**agent**" exploring the search space and move along successor relations, hoping to improve its position in the long run.

Question: Instead of maintaining one agent,
what if we deploy more than one agent?

Population-based Search

A population is a set of candidate solutions

A population-based search conducts search by maintaining a population

Advantages:

- Search diversification
- Combination of promising traits in different candidate solutions

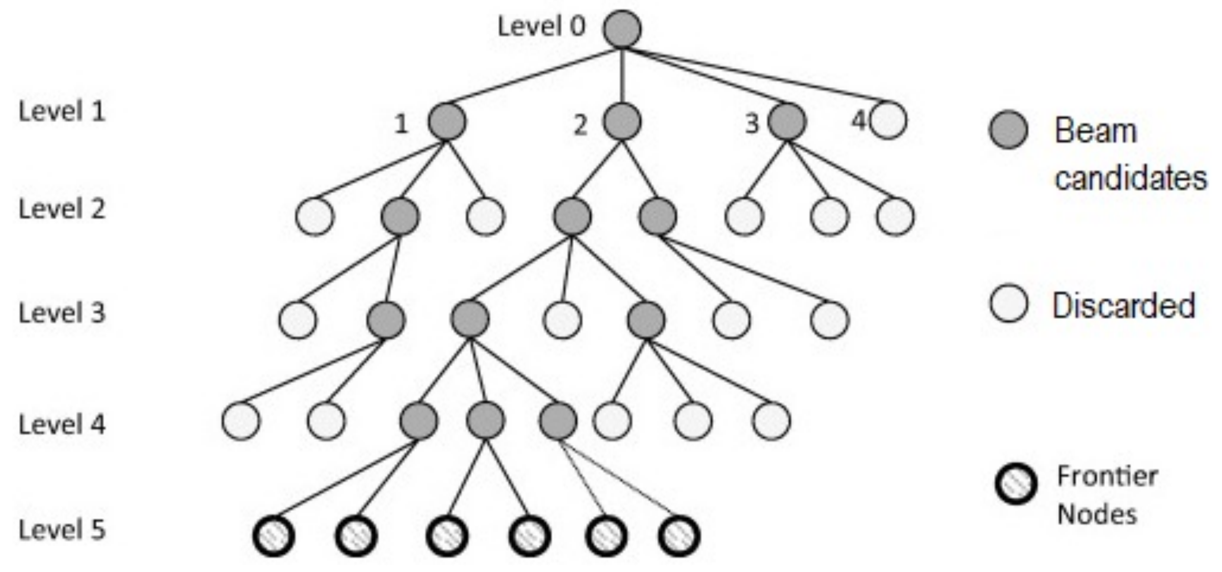
Example population-based search strategies:

- Local beam search
- Genetic algorithm

Population-based Search 1: Local Beam Search

Local beam search changes the Iterative Best Improvement strategy by maintaining a population of size M, where $M \geq 1$ is a parameter.

- Initialise a population $x_1, \dots, x_M$
- At every step, examine all successors of the current population and select the top M successors with the lowest cost
- Replace $x_1, \dots, x_M$ with the selected M successors



What is the difference between greedy descent with M random restarts and local beam search?

Stochastic Beam Search

Disadvantage of local beam search: The search may concentrate on one branch after a few steps.

Stochastic beam search introduces randomness to local beam search, choosing the M successors using Boltzmann distribution. Formally, the probability that a successor y is selected is

$$p(y) \propto e^{-\beta val(y)}$$

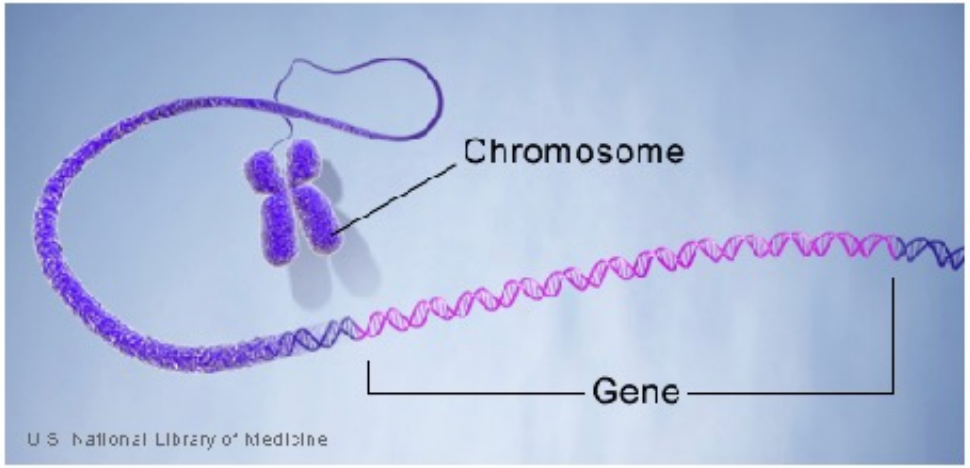
Population-based Search 2: Genetic Algorithms

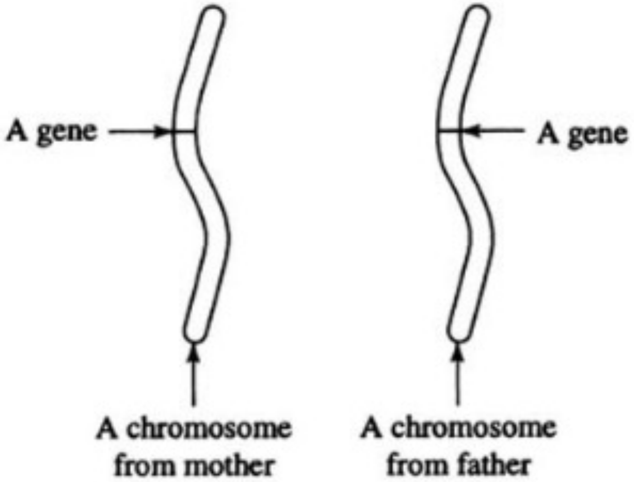
Genetic algorithms are search strategies inspired by genetics and evolution in biology.

- Chromosomes carry genetic information of an organism in the form of genes. They are represented by character strings in base-4 alphabet

A(adenine), C(cytosine), G(guanine), T(thymine)

- An individual carries a number of matching chromosome pairs

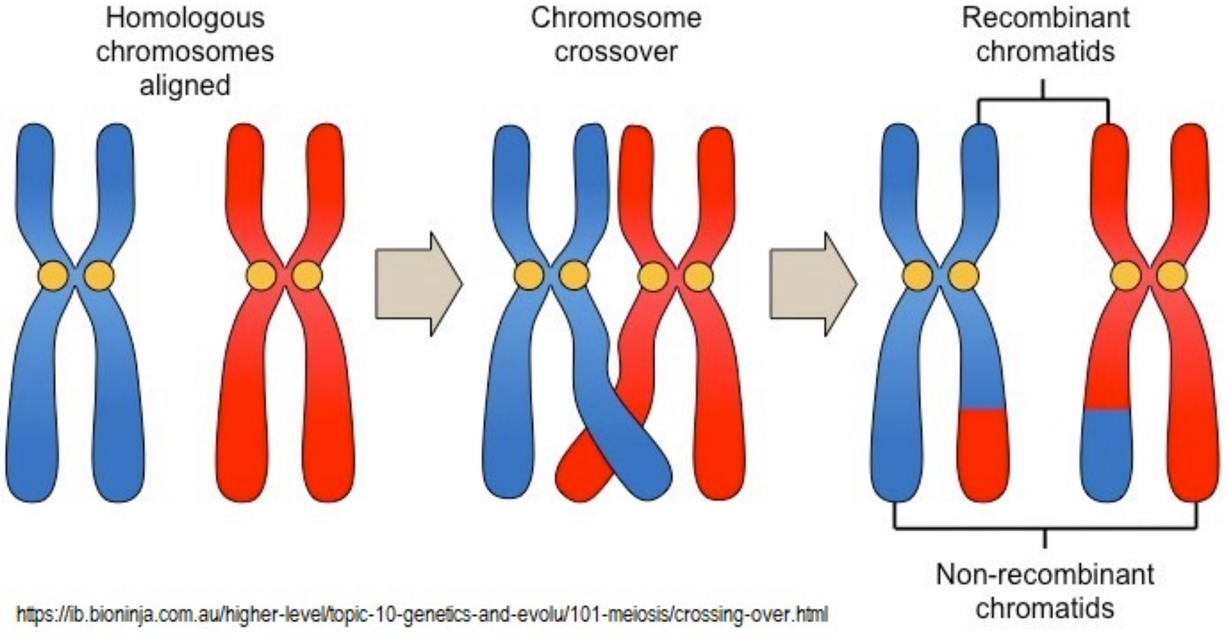




A reproduction process occurs between two individuals, i.e., mother and father.

The chromosomes of these two individuals make a pair of chromosomes for the child:

- One chromosome from each chromosome pair of an individual is selected.
- Chromosomal crossover involves recombination between selected paired chromosomes inherited from each of one's parents.
- Occasionally, some genes value may mutate due to random events



https://ib.bioninja.com.au/higher-level/topic-10-genetics-and-evolu/101-meiosis/crossing-over.html

Question: What is evolution?

Genotype is the collection of genes of an organism

Phenotype is the organism's observable characteristics

- **Fitness** refers to how adapted an organism is to its environment.
- Fitness depends on phenotype and phenotype depends on genotype.
- There is an evolutionary tendency towards phenotype that are more adapted to the environment from generation to generation.
- Therefore, evolution can be considered as a search process for fit phenotypes through iterations of changes to genotypes.

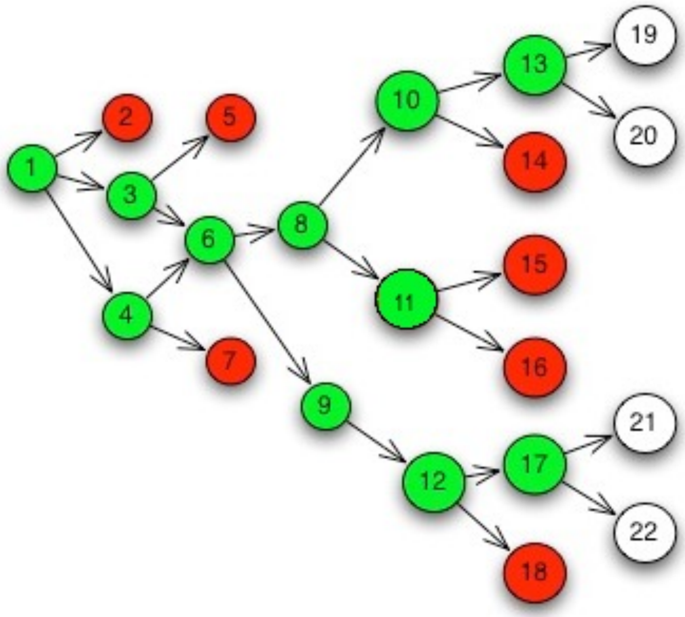
Necessary conditions of evolution:

- A population of individuals
- Variety among individuals
- Differences in ability to survive in the environment that are associated with the variety
- The individuals are able to reproduce
- An offspring inherits features of the parents, with potential mutations.

Evolution as Search

Evolution is continual process in which the following steps are repeated:

1. Reproduce a pool of individuals
2. Select some individuals with high fitness
3. Perform crossover and mutation between selected individuals to generate new individuals



Genetic Algorithms

To apply the evolution paradigm as a search strategy, we need the following:

- **Candidate solution representation**: The most common representation of a candidate solution in S is a (fixed-length) string of characters in a finite alphabet. They are also called chromosomes.
- **Fitness**: The fitness function measures how desirable a candidate solution looks.
- **Crossover operation**: Take two chromosomes $s = s_1 s_2 \dots s_l$, $t = t_1 t_2 \dots t_l$ and crossing site $1 \leq c < l$. The crossover operation returns

$$s_1 \dots s_c t_{c+1} \dots t_l \text{ and } t_1 \dots t_c s_{c+1} \dots s_l$$

- **Mutation operation**: Take a chromosome $s = s_1 s_2 \dots s_l$, the mutation operation returns

$$\hat{s}_1 \hat{s}_2 \dots \hat{s}_i \dots \hat{s}_l$$

where $\hat{s}_i \neq s_i$ with some (small) chance

Genetic algorithm (GA) is a variation of stochastic beam search that generates every "successor" using two candidate solutions.

Genetic Algorithm (GA)

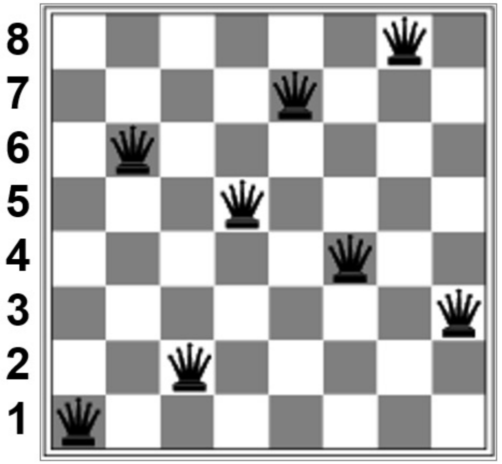
INPUT: search problem, crossover probability p_c, mutation probability p_m, population size M
OUTPUT: candidate solution y

repeat

1. **Reproduction:** Preferentially select from Π a mating pool
2. **Crossover:** Randomly take $x, y \in \Pi$. Do
   With probability p_c: Add a new copy of x and y
   With probability $1 - p_c$: Perform crossover on x and y with random crossing site
   Repeat crossover breeding to generate M new solutions.
3. **Random mutation**: with probability p_m, randomly select a small portion of the solutions and artificially change it.

until Π is stabilised
return the best solution in Π

Example. [8-queens] The chromosome of any candidate solution is a string of length 8 over the alphabet {1,2,…,8}.



String representation
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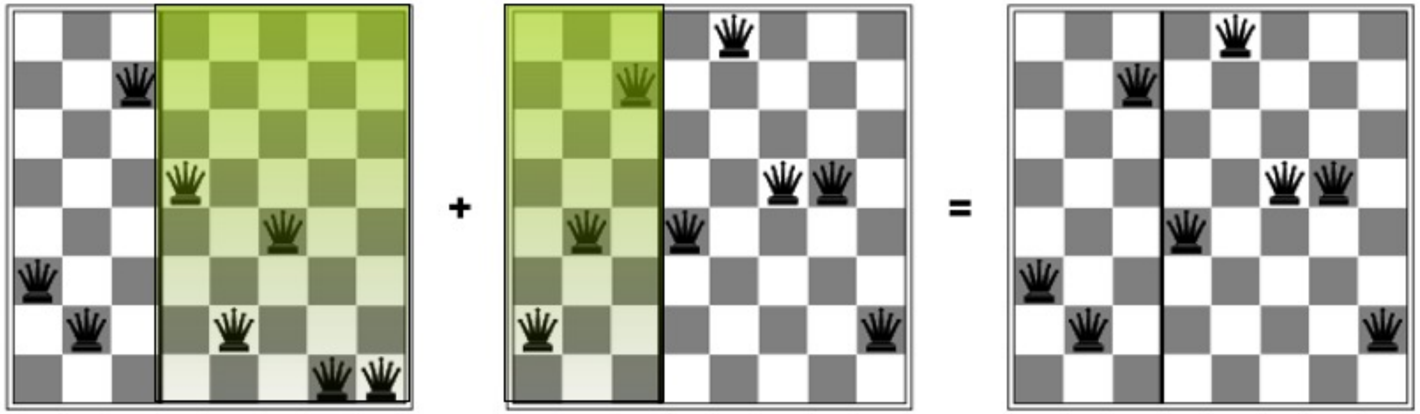
Crossover may help to combine promising partial solutions



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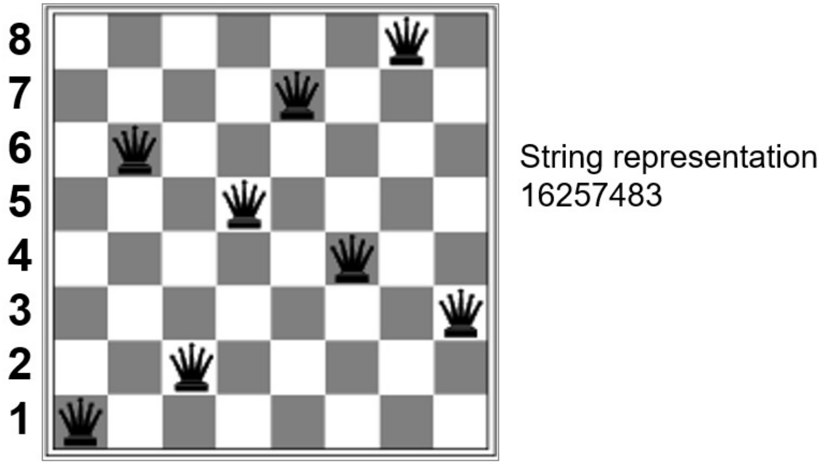
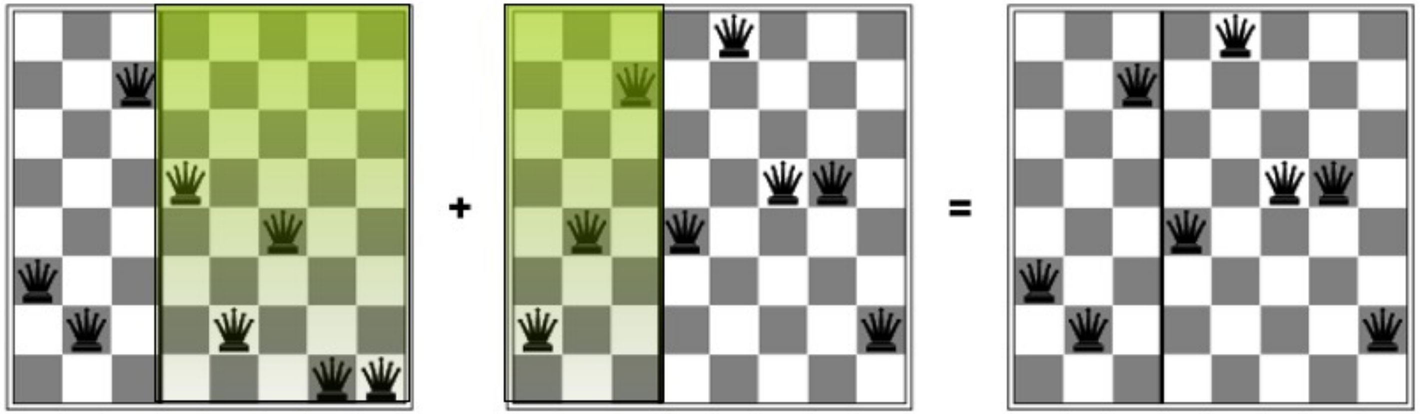


Fitness function $f(x)$: the number of non-attacking pairs of queens



$f(32752411) = 23$ $\quad f(24748552) = 24$ $\quad f(32748552) = 23$

Assume that we adopt the following setting:

- Choose $M = 4$

- Fitness function $f(x)$: the number of non-attacking pairs of queens

  E.g., $f(24748552) = 24$, $f(32752411) = 23$, $f(24415124) = 20$, $f(32543213) = 11$

- When creating mating pool, select a candidate solution using the following probability distribution

$$p(x) = \frac{f(x)}{\sum_{x \in \Pi} f(x)}$$

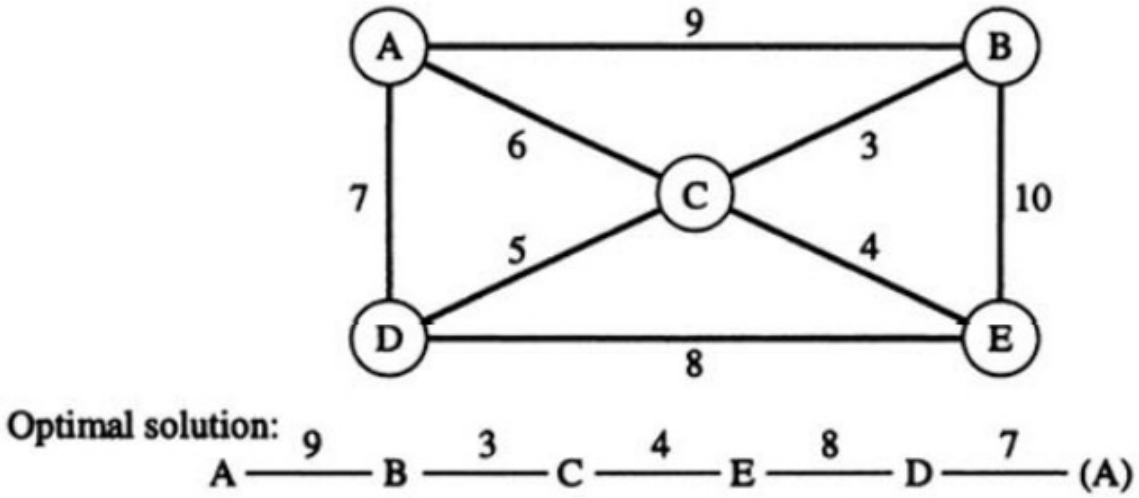
  E.g., $p(24748552) = 31\%, p(32752411) = 29\%, p(24415124) = 26\%, p(32543213) = 14\%$

Run GA:

- Initialise Π: 24748552, 32752411, 24415124, 32543213
- Select mating pool: 24748552, 32752411, 24415124
- Crossover breeding: 32748552, 24752411, 32752124, 24415411
- Mutation: 32748152, 24752411, 32252124, 24415417

1 step of GA leads to a general improvement from the initial population.

Example. [TSP] Consider four cities A,B,C,D,E



The goal is to find a tour with the lowest cost using GA.

Two key questions:

- What are the chromosomes?
- What is the fitness function?

Question 1. (Chromosome) – representing a candidate solution using a string

First attempt. Express permutations as strings.

E.g. ADEBC, ADCEB

However, this representation is not closed under crossover

E.g., ADE|BC, ADC|EB → ADEEB, ADCBC.

A different representation.

- How if we represent the solution as the permutation of a tour?
- Consider the sequence ADEBC of a tour:
  - Initialize a list of {A, B, C, D, E}
  - Remove one character from the list each time, in the order provided by the sequence (e.g., ADEBC) and record its index (position) in the list
  - The remaining character will form a new list, e.g.,
  - Repeat removing the next character until the list is empty
- The index removed at each iteration denotes the character's position in the remaining list. Thus, the sequence of indexes gives the order of removing positions in the remaining lists.

A different representation.

1. Define index of A,B,C,D,E as 0,1,2,3,4
2. Set $i \leftarrow 0$ and define $enc(i) \leftarrow index(\sigma_i)$
3. Remove σ_i from $\{A, B, C, D, E\}$, decrease the index of each letter whose index is bigger than $index(\sigma_i)$ by 1.
4. Set $i \leftarrow i + 1$. Go back to Step 2 until $i = 5$
5. Return $enc(0)enc(1)enc(2)enc(3)enc(4)$ as encoding $enc(\sigma_0\sigma_1\sigma_2\sigma_3\sigma_4)$

Example. A D E B C

Index Table					Next City and Index	Index List
A	B	C	D	E	A	
0	1	2	3	4	0	0
B	C	D	E		D	
0	1	2	3		2	02
B	C	E			E	
0	1	2			2	022
B	C				B	
0	1				0	0220
C					C	
0					0	02200 ←

Facts.

- This encoding method produces a code $\tau_0\tau_1\tau_2\tau_3\tau_4$ where τ_i is a number from $\{0, \dots, 4-i\}$
- Every code $\tau_0\tau_1\tau_2\tau_3\tau_4$ represents a permutation of ABCDE
- This encoding method is closed under the crossover operation.

E.g., $\tau_1 = ADEBC, \tau_2 = ADCEB$

- enc(ADEBC) = 02200, enc(ADCEB) = 02110
- Crossover (022|00, 021|10) → (02210, 02100)
- dec(02210) = ADECB, dec(02100) = ADCBE

Question 2. (Fitness function)

- The fitness function should correlate with total cost of a tour, a higher loss corresponds to lower fitness

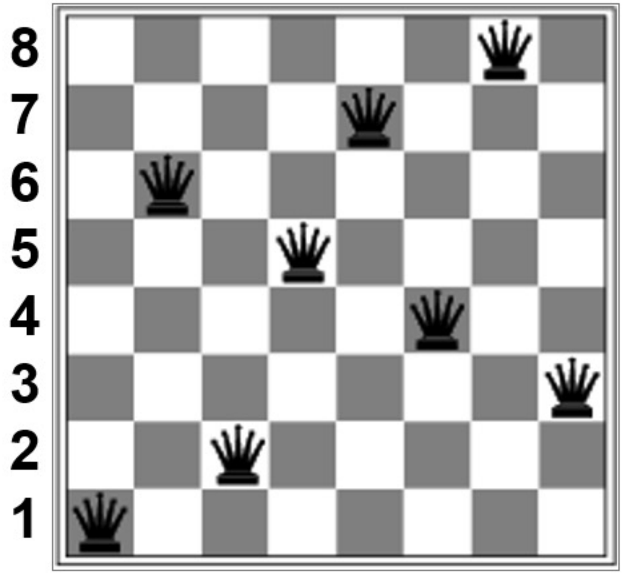
$$\text{E.g., } f(\tau) = \frac{1}{val(\tau)}, f(\tau) = \frac{1}{val(\tau)^2}, f(\tau) = k - val(\tau) \text{ for fixed } k.$$

- If solution quality and fitness correlate too weakly, search becomes inefficient and requires too many steps.
- If solution quality and fitness correlate too strongly, the population easily concentrates and loss in diversity.

Schemata Theorem

How effective is GA in finding good (optimal) solutions?

- Candidate solutions are represented by chromosomes, i.e., fixed length strings over a finite alphabet Σ.
- The traits of a candidate solution can be captured by patterns in the chromosomes.



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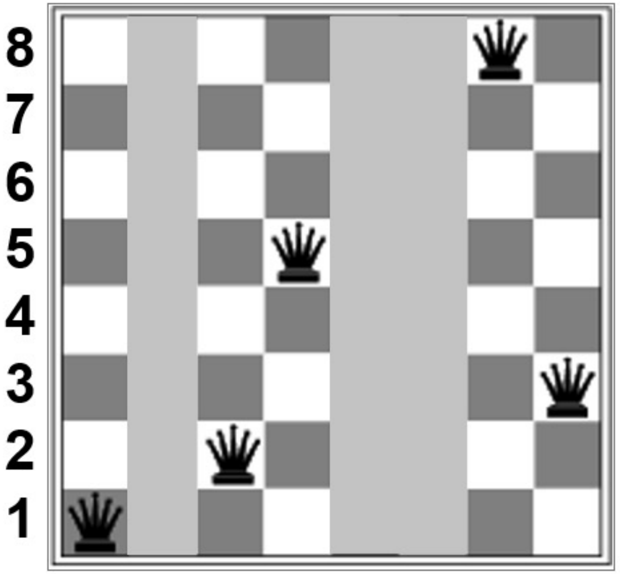

A "trait" 1∗25∗∗83

Definition

For a finite alphabet Σ, a schema is a string over the alphabet $\Sigma \cup \{*\}$, where $*$ is a symbol not in Σ

Example. For alphabet {0,1}, 101*,****,1010 are schemata.

Given a finite alphabet Σ and integer length $L > 0$:

- We say a chromosome $h = s_1 \ldots s_L$ contains a schema $H = \sigma_1 \ldots \sigma_L$ if $s_i = \sigma_i$ whenever $\sigma_i \neq *$. Defined as $h \vDash H$.

  E.g., 1010 contains schemata 1010, *010, 1*10, 10**, *0**, ****, etc.

  Fact. Any length L chromosome contains exactly 2^L schemata.

- The size $|H|$ of scheme H is the number of chromosomes that contains H, i.e., $|H| = |\{h|h \vDash H\}|$.

  E.g., $|101*| = 2, |1010| = 1, |****| = 16$

- The fitness of a schema H, denoted by $f(H)$, is the average fitness of all chromosomes that contain schema H:

$$f(H) = \frac{\sum_{h \vDash H} f(h)}{|H|}$$

- The average fitness $\tilde{f}$ over Σ is $f(* \cdots *)$, i.e., the average fitness of all possible chromosomes.
- The order of a schema H, denoted by $o(H)$, is the number of Σ-positions in H.

  E.g., $o(101\,*) = 3, o(1\,***) = 1$

- The defining length $\delta(H)$ of a schema H is the distance between the last and first Σ-positions in H.

  E.g., $\delta(101\,*) = 3 - 1 = 2, \delta(1 * 01) = 4 - 1 = 3, \delta(1\,***) = 1 - 1 = 0$

- Let $\Pi(t)$ denote the population Π after running t iterations of GA.
- Let $m(H,t)$ denote the number of chromosomes in $\Pi(t)$ that contain H.

Example. If $\Pi(t) = \{1010,1011,0000,0001\}$,

- $m(**01, t) = 1$,
- $m(*0**, t) = 4$.

Schemata Theorem (Fundamental Theorem of GA)

Suppose GA runs on chromosomes of length L with crossover probability p_c and mutation probability p_m. For any schema H,

$$m(H, t+1) \geq m(H,t) \cdot \frac{f(H)}{\tilde{f}}\left[1 - \delta(H)\frac{p_c}{L-1} - o(H)p_m\right]$$

$$m(H, t+1) \geq m(H, t) \cdot \frac{f(H)}{\tilde{f}} \left[1 - \delta(H) \frac{p_c}{L-1} - o(H) p_m \right]$$

relative fitness of H

effect of crossover

effect of mutation

Schemata theorem tells us how and why certain “trait” survives in population Π:

- Short traits with high average fitness has a high chance of survival
- If $f(H) < \tilde{f}$, the right hand side converges to 0.

Summary of the topic

The following are the main knowledge points covered:

- Population-based search
  - Local beam search
  - Stochastic beam search
  - Genetic algorithms
    - Chromosomes: representations of candidate solutions that determines fitness
    - Fitness: determines how a candidate solution is likely to survive
    - Crossovers and mutations
    - The schemata theorem